SEAT NO:__________

No. of Printed Pages: 02

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SARDAR PATEL UNIVERSITY

BCA (Sem-IV) Examination – April 2024

US04CBCA54: Operating System

Date: 1st April 2024 (Monday)

Time: 2.00pm to 5.00pm **Total Marks: 70**

Note: Figure indicate at right, is maximum marks for each question.

Q:1 Select an appropriate option for the following statement **[10]**

1. Time- Slice Scheduling algorithm is also called ________
   A. FCFS B. SJF C. Round Robin D. Priority

2. PCB Stands for ______
   A. Program Control Block B. Process Control Block
   C. Process Communication Block D. Program Communication Block

3. Which one of the following is the operating system service?
   A. Accounting B. Error Detection C. Communication D. All of these

4. ________ page replacement algorithm gives minimum Page Faults.
   A. FIFO B. Optimal C. LRU D. None of these

5. Demand paging is _________.
   A. Paging with swapping B. Paging without swapping
   C. Swapping with fragmentation D. Swapping without fragmentation

6. _______ is not a valid memory allocation algorithm
   A. Suit-fit B. Worse-fit C. Next-fit D. First-fit

7. Printer is an example of _______ resource
   A. Physical B. Logical C. Virtual D. None of these

8. _________ is not a valid deadlock characteristic.
   A. Hold & Wait B. Pre-emption C. No-Preemption D. All of these

9. Which command is used to get help for specific other command/s?
   A. man B. help C. cmdhelp D. ls

10. In shell script the if statement ends with ______.
    A. end B. fi C. endif D. None of these

Q:2 Give brief answer for the following question (Attempt any Ten) **[20]**

1. Give any three examples of Real-Time operating system.
2. Draw diagram of PCB.
3. List key elements of operating system.
4. What is Belady's anomaly?

[P.T.O]

5. What is compaction? What is the use of compaction?
6. Discuss best-fit memory allocation technique.
7. When does Race-condition arise?
8. Why is Linux an open source?
9. Explain concept of resource utilization.
10. What is shell script?
11. Explain use of echo command.
12. Discuss 'expr' command of Linux.

Q:3 [A] Define: Operating System. Explain services of an operating system. **[05]**

[B] Explain process states and its transition. **[05]**

OR

[A] Discuss about multiprogramming operating system with its merits and demerits. **[05]**

[B] List structures of operating system. Explain Microkernel in brief. **[05]**

Q:4 [A] What is Fragmentation? Discuss types of fragmentation. **[05]**

[B] Discuss Optimal Page Replacement Algorithm and Calculate Page faults using Optimal page replacement algorithm for following reference string: (Number of Frames = 3)
Reference string = 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1 **[05]**

OR

[A] Discuss concept of swapping. **[05]**

[B] What is Page Fault? Discuss page fault handling technique. **[05]**

Q:5 [A] Explain Critical-Section problem in details. **[05]**

[B] Explain Algorithm-1 for two-process solution. **[05]**

OR

[A] What is deadlock? Explain all necessary conditions for occurrence of deadlock? **[05]**

[B] Discuss Algorithm-3 for two process solution. **[05]**

Q:6 [A] Discuss while and for loop of Linux with example. **[05]**

[B] Explain File Redirection Operators with example. **[05]**

OR

[A] Discuss case statement with example. **[05]**

[B] Write a note on File Accessing Permission (FAP). **[05]**

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